Beyond Parameter Arithmetic: Sparse Complementary Fusion for Distribution-Aware Model Merging

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Abstract

Model merging combines specialized large language models directly in weight space without additional training. In our evaluated configurations, however, several merging baselines exhibit generation dysregulation, including repetition, termination failure, circular reasoning, and semantic collapse. At 14B/32B scales, their length-based generation dysregulation rate (L-GDR), an abnormal-length diagnostic, exceeds 97% on GSM8K, alongside accuracy losses of up to 70 percentage points. We propose Sparse Complementary Fusion with Reverse KL (**SCF-RKL**), a parameter-only merging framework that selects complementary parameters using reverse KL divergence between weight-space proxy distributions, given a user-specified anchor order. Proxy-space entropy and subspace bounds motivate its sparse binary updates. Experiments on 7B–32B language-model pairs and an eight-model vision composition show a competitive capability–stability trade-off, with near-zero L-GDR under the reported reference protocols. The analysis does not provide an end-to-end generation guarantee.

1 Introduction

Model merging has emerged as a compelling paradigm for composing specialized large language models (LLMs) directly in weight space [1–4], offering a training-free route to multi-capability systems built from open-source or custom-trained specialists [5–7]. Yet a critical failure mode has gone largely unnoticed: **many evaluated merging methods induce generation dysregulation—a spectrum of failure modes ranging from structural repetition and termination failure to circular reasoning, and at the extreme, full semantic collapse into incoherent symbols (Figure 1, Right)—even when source models are nearly free of such behavior**. As shown in Figure 1 (Left), two patterns occur in the reported model families under the reference protocols: **(Suppression)** when one source model already exhibits high dysregulation (e.g., 7B and 8B), several methods propagate it above 97% while SCF-RKL is below 1%; **(Immunity)** when both source models are clean (14B/32B: $<$0.15%), the evaluated dense baselines reach 97–100% while SCF-RKL remains near zero. In multi-turn dialogue and long-horizon reasoning, such responses can exhaust the context window and disrupt subsequent generation.

Empirical failure pattern. Our ablations associate these failures with dense parameter-space averaging: it introduces *phantom parameters* (values absent from both parents), lacks structural guidance in parameter selection, and perturbs the anchor's generative control mechanisms at scale (§2). This motivates a design challenge: **how can model merging integrate complementary capabilities while preserving generation stability?**

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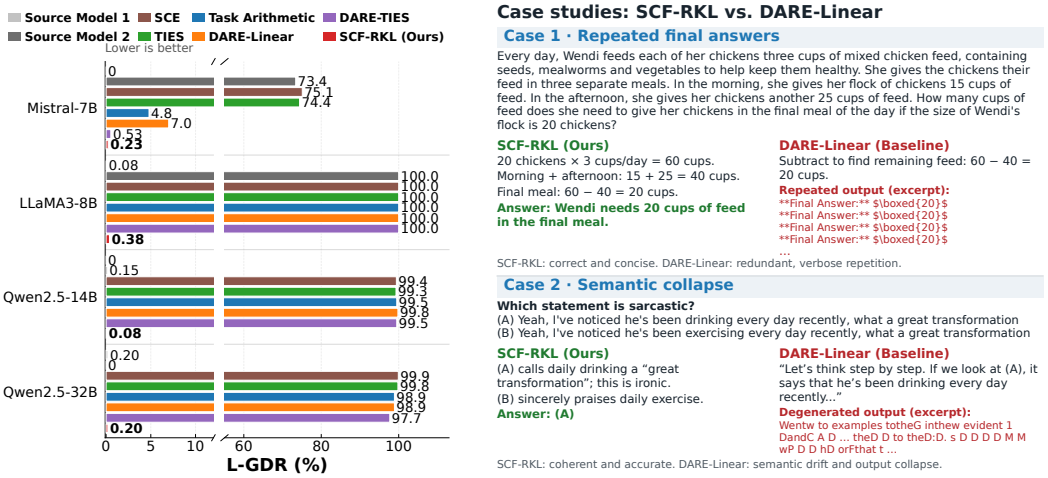


Figure 1: **(Left)** GSM8K L-GDR across four model pairs under the original reference protocols (broken x-axis). **(Right)** Summarized reasoning and failure excerpts: DARE-Linear repeats answers (*top*) or produces semantic collapse (*bottom*); SCF-RKL answers correctly and terminates normally.

Our approach. We introduce **Sparse Complementary Fusion with Reverse KL (SCF-RKL)**, a distribution-aware, parameter-only merging framework whose operation requires no data once the anchor order is specified. Instead of blending all parameters, SCF-RKL selects a sparse set of the most *structurally informative* updates from the complementary model, guided by reverse Kullback–Leibler divergence [8] computed on parameter-group proxy distributions. In the audited Mistral-7B run, the realized global replacement rate is 5.4261% (5.0481–6.6031% across transformer layers); the rate is model-pair- and tensor-dependent rather than a fixed universal constant. Reverse KL weights proxy discrepancies by the anchor distribution, providing an anchor-dependent criterion for allocating updates across rows. Under the tested reference protocols, SCF-RKL combines competitive capability with low L-GDR (Figure 1); we also evaluate reasoning–safety fusion.

Contributions.

1. **Failure mode identification.** We document generation dysregulation in the evaluated merging settings—manifesting as structural repetition, termination failure, circular reasoning, and semantic collapse—and use controlled comparisons to motivate binary selection, low-density updates, and anchor-centric saliency.
2. **SCF-RKL method.** An effective sparse fusion framework using reverse KL divergence as an information-theoretic saliency criterion, with a parameter-only implementation once the anchor order is supplied.
3. **Theoretical analysis.** We give proxy-space entropy and subspace bounds that apply to sparse binary masks (Proposition 3.1, Theorem 3.2), use a local Fisher expansion only as motivation for the saliency, and evaluate the resulting criterion with controlled ablations (§3.4).
4. **Empirical evaluation.** We study four fusion scenarios: similar-capability pairs, strong–weak pairs, reasoning–safety fusion, and multi-model vision composition.

2 Sparse Complementary Fusion with Reverse KL

Motivation and metric scope. We use the ***length-based generation dysregulation rate*** (L-GDR) as a screen for unusually long completions, not as a semantic-collapse label. For each benchmark/source-model pair, the two unmerged parents are run once with a frozen reference configuration ($T = 0.6$, $n = 4$, and `max_new_tokens` $= 4096$). Let $\mathcal{R}$ contain nonempty parent completions that terminate through EOS or the configured service stop; request failures are excluded from the reference-eligible set and tracked separately from returned-completion metrics. With the model-pair tokenizer, let m be the median number of completion tokens in $\mathcal{R}$ (prompt tokens excluded). For every returned completion r, let $L(r)$ be its completion-token length, including max-token-truncated outputs in the

Table 1: Compact L-GDR robustness check on matched Mistral-7B GSM8K runs. Each cell is SCF-RKL/TSV-Merge [9] (%). Lower L-GDR is better; all nine paired L-GDR-gap confidence intervals favor SCF-RKL.

Temperature	@5×	@10×	@20×	Score gap
0.0	1.36/6.60	1.29/6.37	1.29/6.29	+9.02
0.6	1.76/14.52	0.99/13.32	0.59/12.36	+9.61
1.0	2.96/16.68	0.61/9.63	0.15/4.70	+6.52

denominator. We report

$$\text{L-GDR@}k\times = \frac{1}{N}\sum_{r=1}^{N}\mathbf{1}\{L(r) > km\}, \qquad k \in \{5, 10, 20\}.$$

Here "reference-eligible" describes the stopping rule and does not imply answer correctness. We define the complementary protocol metrics as follows: max-hit is the fraction of returned completions whose finish reason is the generation-length limit; normal-stop is the fraction ending through EOS or the registered service stop; and Rep-4 is the mean token-level fraction of repeated 4-grams, $1 - (\#\text{distinct 4-grams})/(\#\text{all 4-grams})$, computed on completion tokens only. Accuracy among L-GDR hits and a qualitative semantic audit test whether long outputs are valid reasoning. L-GDR is therefore interpreted jointly with task accuracy, max-hit, normal-stop, Rep-4, hit accuracy, and audit; the formal metric table and temperature/budget/presence-penalty sweeps are in Appendix B.4.

Dense parameter-space averaging can introduce *phantom parameters*—values absent from both parents (Proposition 3.3)—and perturb many coordinates simultaneously. These properties motivate testing sparse binary selection with an explicit criterion for choosing updates; they do not by themselves establish the cause of generation failures. Ablation shows that the evaluated binary mask and sparsity choices matter: under the common tensor-wise thresholding protocol, random masking gives 72.3% dysregulation versus 0.5% for SCF-RKL, while changing the threshold multiplier also changes the observed rate (§3.4). This diagnosis motivates three design principles: **(i)** *binary selection over interpolation*—eliminating phantom parameters entirely; **(ii)** *low-density binary updates*—bounding perturbation to preserve the anchor's dominant structure (Theorem 3.2); **(iii)** *anchor-centric saliency*—prioritizing updates using discrepancies between the parents' row-wise proxies. SCF-RKL instantiates all three in a single parameter-only framework once the anchor order is supplied.

2.1 Problem formulation

Let $\theta_a \in \mathbb{R}^d$ denote the parameters of an anchor (source) model and $\theta_c \in \mathbb{R}^d$ those of a complementary model trained or specialized under different objectives (e.g., reasoning, instruction-following, or safety alignment). Our goal is to construct a fused model θ_f that selectively incorporates the strengths of θ_c while maintaining the stable generative behavior of θ_a.

We define the fused parameters as:

$$\theta_f = \theta_a + M \odot (\theta_c - \theta_a) \tag{1}$$

where $M \in \{0, 1\}^d$ is a sparse fusion mask that determines which parameters are updated. Because M is binary, every coordinate of θ_f is drawn directly from either θ_a or θ_c—no interpolated values are introduced.

Anchor selection and directional asymmetry. The experiments used a benchmark-informed anchor order, specified before merging; this is separate from the parameter-only merge operation. The audited Mistral-7B run retains 94.5739% of anchor parameters. Direction-swap results support an aggregate heuristic, but near-tied pairs show task-level reversals (Appendix B.7). In deployment, users can choose the anchor by the behavior, alignment policy, interface, or provenance to preserve. Automatic ordering without such prior information remains open. RKL weighting (§2.2) allocates updates across rows.

The **central challenge** is to design $M \in \{0, 1\}^d$ to inject only **informationally meaningful** updates while avoiding large-scale parameter drift that causes **generation dysregulation**.

2.2 Information-theoretic saliency via reverse KL divergence

Parameter-group proxy distributions. For each parameter group (e.g., per output channel or hidden dimension), we construct a *proxy distribution* by applying softmax row-wise: $p_i = \mathrm{softmax}(\theta_{ci,:})$ and $q_i = \mathrm{softmax}(\theta_{ai,:})$, where $p_i, q_i \in \Delta^{d_{\text{in}}-1}$ are deterministic functions of the model parameters, defined independently for each output dimension i. Normalizing each row via softmax maps the parameter vector to a probability simplex, enabling information-geometric comparison and capturing relative structural shifts between models beyond raw magnitude differences.

Reverse KL as an information-sensitive metric. We measure the structural discrepancy between proxy distributions in weight space using the **reverse Kullback–Leibler divergence**:

$$\mathrm{RKL}(q_i \parallel p_i) = \sum_j q_{i,j} \log \frac{q_{i,j}}{p_{i,j}} \tag{2}$$

The weighting distribution is the anchor proxy q. The resulting divergence is a scalar for each row: it reweights rows, while coordinate magnitudes determine the ordering within a row. This construction does not guarantee preservation of the anchor's output distribution.

Information-theoretic saliency measure. Guided by the reverse KL divergence, we compute an information-theoretic saliency score for parameter selection:

$$I_{i,j} = |\theta_{c,i,j} - \theta_{a,i,j}| \, \mathrm{RKL}(q_i \parallel p_i) \tag{3}$$

This formulation couples **parameter deviation magnitude** with **information-geometric impact**; in the evaluated comparisons, this downweights large updates from rows with little proxy divergence relative to magnitude-only selection.

2.3 Dynamic sparse complementary fusion

To determine which parameters should be updated, we adopt **distribution-adaptive thresholding**.

For each weight tensor, let I denote its saliency scores. We compute their median $\mathrm{med}(I)$ and interquartile range $\mathrm{IQR}(I) = Q_{0.75}(I) - Q_{0.25}(I)$. The fusion threshold is defined as:

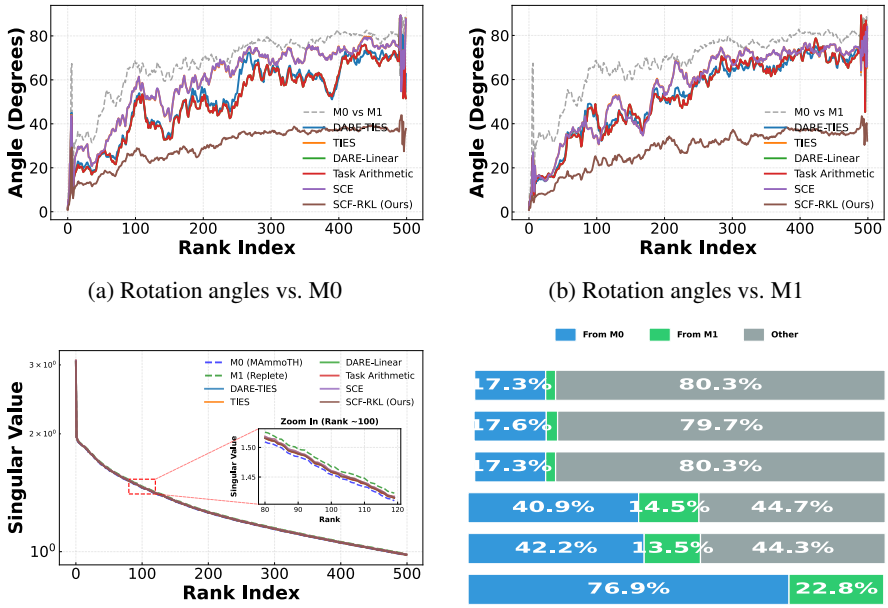
$$\tau = \mathrm{med}(I) + \alpha \cdot \mathrm{IQR}(I) \tag{4}$$

where α is a scale factor of IQR. The fusion mask is then $M_i = \mathbb{I}[I_i \geq \tau]$, selecting parameters whose information-theoretic saliency scores lie in the upper statistical tail. This yields a sparse update that adapts automatically across layers, model scales, and architectures. This Tukey-inspired IQR threshold requires no held-out calibration set. Unlike the classical upper fence, it is centered at the median, not $Q_{0.75}$. In the Mistral-7B ablation, performance is non-monotonic over $\alpha \in [1.0, 2.0]$, peaking at $\alpha = 1.5$; the observed range is 4.22 percentage points (§3.4). This is a sensitivity range rather than an invariance claim; $\alpha = 1.5$ is the fixed setting used in the main comparisons.

Algorithm 1: Sparse Complementary Fusion with Reverse-KL Sparsification

Input: anchor params θ_a, complementary params θ_c, smoothing $\epsilon = 10^{-8}$,
quantiles $q_{\text{low}} = 0.25$, $q_{\text{high}} = 0.75$, $q_{\text{center}} = 0.5$,
IQR scale $\alpha = 1.5$
Output: fused parameters θ_f
$\Delta \leftarrow |\theta_c - \theta_a|$
$q \leftarrow \mathrm{softmax}(\theta_a) + \epsilon$; $p \leftarrow \mathrm{softmax}(\theta_c) + \epsilon$
$D_{\mathrm{KL}}^{\mathrm{rev}} \leftarrow \sum_i q_i \log(q_i/p_i)$ *(per output dim)*
Broadcast $D_{\mathrm{KL}}^{\mathrm{rev}}$ to match shape of Δ
$\mathcal{I} \leftarrow \Delta \odot D_{\mathrm{KL}}^{\mathrm{rev}}$ *(importance scores)*
Flatten $\mathcal{I}$; compute quantiles Q1, Q3, μ
$\tau \leftarrow \mu + \alpha \cdot (\mathrm{Q3} - \mathrm{Q1})$ *(dynamic threshold)*
$M \leftarrow \mathbf{1}[\mathcal{I} \geq \tau]$ *(binary mask)*
$\theta_f \leftarrow \theta_a + M \odot (\theta_c - \theta_a)$
return θ_f

Computational complexity. SCF-RKL is offline, one-shot, and training-free. For L layers with T matrix types per layer and weight matrices $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, the implementation computes exact order statistics tensor by tensor. For a tensor with d_T entries this costs $O(d_T \log d_T)$ time and $O(d_T)$ auxiliary space; the total is the sum over tensors (and is upper-bounded by $O(d \log d)$ for d total parameters). On H800 (Table 27), the measured time and peak GPU memory are 59 s / 4.2 GB (7B), 331 s / 24.7 GB (14B), and 352 s / 24.7 GB (32B). This is approximately 1.6–2.2× the wall-clock time of Task Arithmetic, incurred once at merge time.



(a) Rotation angles vs. M0

(b) Rotation angles vs. M1

(c) Singular value spectrum comparison

(d) Parameter source distribution

Figure 2: **Geometric and spectral analysis of fusion methods.** **(a)** Principal angle rotation vs. M0 at layer 15. **(b)** Principal angle rotation vs. M1 at layer 15. Gray dashed line: gap between parent models M0 and M1. **(c)** Singular value spectrum; inset zooms around rank 100. **(d)** Parameter source distribution across six fusion methods (top to bottom: DARE-Linear, DARE-TIES, Task Arithmetic, TIES, SCE, SCF-RKL (Ours)). SCF-RKL maintains rotation below $40°$ (vs. $\sim 80°$ for dense methods), spectral interpolation between parent models, and NSS $\approx 10^{-3}$.

3 Proxy-space analysis and empirical diagnostics

This section provides a scoped motivation and diagnostics for the selection rule. We first define the weight-space proxy, then state the local approximation behind the saliency, give mask-agnostic sparse-fusion bounds, and finally compare the proxy with intermediate-layer logits and controlled ablations. These steps do not constitute an end-to-end guarantee for autoregressive generation. All distributions below live in *proxy space*, where $q_i = \mathrm{softmax}(\theta_{a,i,:})$ and $p_i = \mathrm{softmax}(\theta_{c,i,:})$ are row-wise normalizations of weights; they are not token distributions.

3.1 Saliency design: a Fisher-inspired heuristic

Given a fixed sparsity budget, which coordinates should be selected? A second-order expansion of this *weight-space* proxy divergence with a diagonal Fisher approximation motivates, but does not derive exactly, the following score (Appendix A.2, A.4):

$$\mathcal{I}_{i,j} = |\Delta_{i,j}| \cdot D_{\mathrm{KL}}^{\mathrm{rev}}(q_i \| p_i) \tag{5}$$

The score has two transparent roles: $|\Delta_{i,j}|$ ranks coordinates within a row, while $D_{\mathrm{KL}}^{\mathrm{rev}}(q_i \| p_i)$ re-weights rows. The Fisher expansion is valid only for small $\|\Delta_i\|_2$, whereas SCF-RKL deliberately selects a high-saliency tail. We therefore use the expansion as heuristic motivation and test the resulting criterion empirically rather than treating it as a quantitative guarantee.

3.2 Formal properties of sparse fusion

We record three formal properties of sparse binary fusion. They hold for any mask satisfying the stated assumptions; they are not exclusive to RKL. RKL only determines which coordinates receive the available budget.

Entropy preservation. Assume the anchor and fused weight rows, interpreted as proxy logits, lie in $[-C, C]^{d_{\mathrm{in}}}$. On this domain, $H \circ \mathrm{softmax}$ is L-Lipschitz in the ℓ_1 norm, with $L = 2C$

(Appendix A.5). This is a weight-space assumption; LayerNorm/RMSNorm does not impose it on raw weights.

Proposition 3.1 (Entropy Preservation Bound)**.** *Let $\mathcal{S}_i = \{j : M_{i,j} = 1\}$ and $q_{f,i} = \mathrm{softmax}(\theta_{f,i,:})$. Under the assumption above,*

$$H(q_{f,i}) \;\geq\; H(q_i) - L \cdot \sum_{j \in \mathcal{S}_i} |\Delta_{i,j}| \tag{6}$$

Masking cannot increase the perturbation's ℓ_1 norm. Sparsity alone, however, does not guarantee a non-vacuous entropy bound.

This bound holds for any mask. In our experiments, RKL allocates selections to rows with larger proxy divergence and is empirically useful among the evaluated criteria: magnitude selection reaches 41.52 average accuracy versus 43.19 for RKL (Figure 4).

Geometric stability.

Theorem 3.2 (Bound on Subspace Rotation)**.** *Let $W_a, W_c \in \mathbb{R}^{m \times n}$, $\delta = \sigma_r(W_a) - \sigma_{r+1}(W_a) > 0$, and $E = M \odot (W_c - W_a)$ with $\|E\|_2 < \delta/2$. For the leading r left singular subspaces U and $\tilde{U}$ of W_a and $W_a + E$, a conservative consequence of Wedin's theorem and Weyl's inequality [10, 11] is*

$$\sin \Theta_{\max}(U, \tilde{U}) \leq \frac{\sqrt{2}\,\|E\|_2}{\delta - \|E\|_2}. \tag{7}$$

The same bound holds for the right singular subspaces. The reported Mistral-7B ratio $\|E\|_2/\delta < 0.3$ at $\alpha = 1.5$ satisfies the assumption (Appendix A.8). The bound depends on perturbation magnitude and spectral separation, not on density alone.

Proposition 3.3 (No Phantom Parameters)**.** *For binary mask $M \in \{0,1\}^d$, every coordinate satisfies $\theta_{f,i} \in \{\theta_{a,i}, \theta_{c,i}\}$. In contrast, any convex combination $\theta_f = \lambda\theta_a + (1-\lambda)\theta_c$ with $\lambda \in (0,1)$ introduces values absent from both parents on all coordinates where $\theta_{a,i} \neq \theta_{c,i}$.*

Binary masking eliminates phantom parameters. The ablation in §3.4 nevertheless shows that masking alone is insufficient in this evaluated setting: the selection criterion changes both capability and observed dysregulation.

Geometric analysis (Figure 2) confirms: SCF-RKL maintains rotation below $40°$ while dense methods approach $80°$; singular values stay bounded between parent models while baselines overshoot; NSS $\approx 10^{-3}$, nearly an order of magnitude below baselines.

3.3 Proxy-space signal validation at intermediate layers

The preceding analysis operates in proxy space. For the saliency criterion to be meaningful, proxy-space divergence should track *intermediate-layer logit-level* divergence—a measurable signal accumulating toward next-token predictions at each layer.

Logit perturbation bound. For any sparse mask, Hölder's inequality gives: for any $h \in \mathbb{R}^{d_{\text{in}}}$ with $\|h\|_\infty \leq B$, $|\theta_{f,i,:} \cdot h - \theta_{a,i,:} \cdot h| \leq B \cdot \|M_i \odot (\theta_c - \theta_a)_{i,:}\|_1$. This bound is mask-agnostic; it bounds a linear projection for a fixed input and is no larger than the corresponding full-replacement bound (since $\|M_i \odot \Delta_i\|_1 \leq \|\Delta_i\|_1$).

Saliency–logit correlation. We compute per-matrix-type Spearman ρ between row-level parameter-space RKL per layer and logit-level RKL $y_\ell = \frac{1}{N}\sum_n D_{\text{KL}}^{\text{rev}}\big(\mathrm{softmax}(W_{\text{lm}} h_{a,\ell}^{(n)}) \,\big\|\, \mathrm{softmax}(W_{\text{lm}} h_{c,\ell}^{(n)})\big)$ from GSM8K prompts.

Results (Figure 3). All seven projection types yield positive ρ (≥ 0.59, all $p < 10^{-3}$; full values in Table 3). This is supporting evidence that the proxy tracks an intermediate-layer signal in the evaluated models, not a causal proof about final-token distributions or generation outcomes. The Hölder inequality gives a mask-agnostic perturbation bound, and the ablation below tests whether the criterion is empirically useful among the evaluated alternatives.

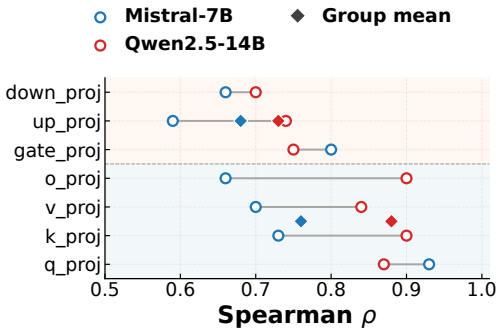


Figure 3: Proxy–logit RKL correlation (Mistral-7B/Qwen2.5-14B): projection-type Spearman ρ (circles) and Attn/MLP means (diamonds); all $p < 10^{-3}$.

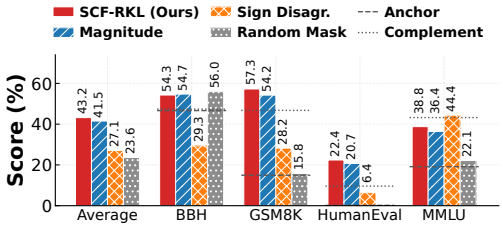


Figure 4: Saliency ablation at α=1.5 with shared tensor-wise IQR thresholding (densities: Appendix B.5). Dashed/dotted: MathCoder2 anchor/Mistral-Ins complement. SCF-RKL has the best mean; per-contrast analysis: §3.4.

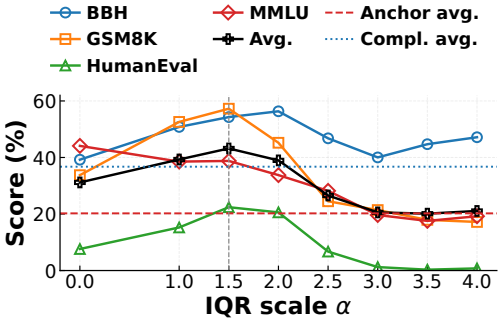


Figure 5: Effect of IQR scale α (Mistral-7B pair). Larger α selects fewer parameters; the highest observed average is at $\alpha = 1.5$. Dashed lines mark each parent model.

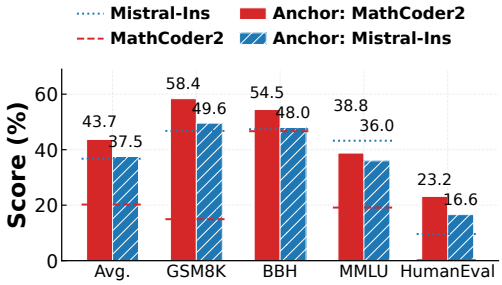


Figure 6: Anchor selection ablation (Mistral-7B pair, $\alpha = 1.5$). The lower-average anchor performs better in this evaluated direction swap.

3.4 Controlled evidence for the selection criterion

We use two controlled ablations to assess the criterion and sparsity setting. They are intervention-style comparisons within this evaluation, not an end-to-end causal identification of generation behavior.

Ablation 1: selection criterion under a shared IQR protocol. At α=1.5, RKL achieves 43.19 average accuracy versus Random 23.63, Magnitude 41.52, and Sign 27.09 (Figure 4). The criteria use the same tensor-wise IQR construction, but their realized densities are not exactly matched (RKL 5.4261%, magnitude 5.9844%, forward KL 8.3863%, and JS 7.6132% in the audited 7B run; Appendix B.5). Random collapses on generation-heavy tasks (GSM8K: 15.81 vs. 57.26; HumanEval: 0.61 vs. 22.41). Thus this comparison is consistent with the selection criterion contributing beyond binary masking under the common protocol, while not isolating a criterion effect at an exactly fixed density. Dysregulation rates mirror this: RKL 0.5%, Magnitude 1.0%, Sign 4.7%, Random 72.3% (Table 22). Thus, the random-mask instantiation of binary masking does not reproduce the same capability–stability trade-off in this evaluated comparison; structured, anchor-centric RKL weighting is empirically useful among the tested criteria for the capability–stability trade-off.

Ablation 2: sparsity level and recovery toward anchor behavior (fixed criterion). Fixing RKL and varying α (Figure 5), dysregulation is minimized at α=1.5 (0.5%) where accuracy peaks (43.19). Beyond $\alpha \geq 2.5$, over-sparsity strips complement contributions and the fused model monotonically recovers the anchor's dysregulation profile: 34.9% $\rightarrow$ 53.6% $\rightarrow$ 69.4% at α=2.5, 3, 3.5 (Table 22). This is consistent with a relationship between the realized sparsity level and the observed dysregulation in this ablation, but does not identify an end-to-end mechanism.

Joint interpretation. These controlled Mistral-7B ablations support RKL selection with IQR-bounded sparsity, without establishing a universal condition or end-to-end guarantee.

4 Experimental setup

Models Experiments span three model families (Mistral, LLaMA 3, Qwen2.5) at 7B–32B scale. We fuse Mistral-7B-Instruct-v0.2[12] with MathCoder2-Mistral-7B[13], Meta-Llama-3-8B-Instruct[14] with MAmmoTH2-8B-Plus[15], and DeepSeek-R1-Distill-Qwen2.5 models (14B/32B) with TinyR1-Preview variants for math, code, or science. Vision uses ViT-B/32[16] checkpoints in the reported eight-task recursive composition.

Fusion baselines We compare with Task Arithmetic[1], DARE-Linear[17], TIES-Merging[2], DARE-TIES[2, 17], and SCE[3], all under standard configurations without fine-tuning. We additionally evaluate four recent parameter-only methods—TSV-Merge[9], ISO-CTS[18], DELLA-Merging[19], and Model Breadcrumbs[20]—on the Mistral-7B and LLaMA3-8B source pairs. These use fixed, disclosed implementation settings without pair-specific tuning; their selected-pair coverage is reported separately from the primary multi-benchmark comparison in Appendix B.8. AdaMerging and LED-Merging are not included in this parameter-only comparison because they require auxiliary unlabeled or labeled task data; PCB-Merging is discussed but remains unevaluated.

Benchmarks Evaluation spans five categories: advanced reasoning (AIME24, AIME25[21], GPQA[22], LiveCodeBench v5[23]), general reasoning (GSM8K, BBH, MMLU), code (HumanEval), instruction following (IFEval[24], IFBench[25]), safety (S-Eval[26], HarmBench[27], JBB-Behaviors[28], WildJailbreak[29], StrongREJECT), and vision (SUN397[30], Cars[31], RESISC45[32], DTD[33], GTSRB[34], MNIST[35], SVHN[36], EuroSAT[37]).

Hyperparameters For SCF-RKL, we use default hyperparameters using a Tukey-inspired IQR threshold: $\epsilon = 10^{-8}$, $q_{\text{low}} = 0.25$, $q_{\text{high}} = 0.75$, $q_{\text{center}} = 0.5$, and $\alpha = 1.5$. In the audited Mistral-7B run this gives 5.4261% global replacement and 5.0481–6.6031% across transformer layers; realized density varies with the tensor and model pair. Sensitivity analysis is in §3.4 (Figure 5); per-benchmark values are in Appendix A.10.

Anchor selection The LLM experiments used an anchor-ordering heuristic based on the original benchmark summaries; this choice is separate from the parameter-only merging operation (§2.1). In the Mistral-7B direction swap (Figure 6), anchoring on MathCoder2 (avg. 20.23) yields 43.70 versus 37.54 in reverse. Near-tied pairs can reverse by task, so the heuristic is not a universal ordering rule. Appendix B.7 distinguishes this archived direction audit from corrected table averages.

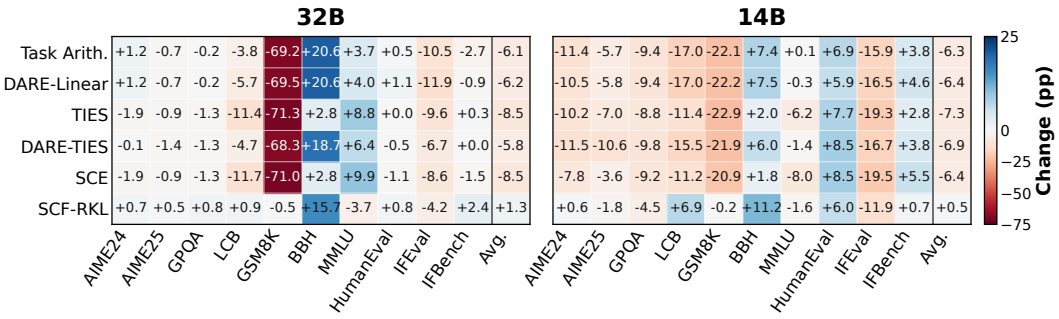


Figure 7: Per-benchmark accuracy change of fused models, measured as the difference between the fused model's score and the average score of the source models. Left: 32B scale. Right: 14B scale.

5 Results of experiments

Figure 7 shows per-benchmark accuracy changes at 32B and 14B scales. At both scales, SCF-RKL **achieves positive gains on the most benchmarks** and is the **only method with a positive average improvement**—all baselines degrade overall. At 7B/8B (Figure 8), SCF-RKL achieves the largest gain on LLaMA-3-8B and second-largest on Mistral-7B, and pass@n scores remain on par with or above source models across all scales (Appendix, Figures 14 to 16).

32B scale. SCF-RKL fuses TinyR1-Preview-Math-32B and TinyR1-Preview-Science-32B, achieving **positive gains on 7 of 10 benchmarks**—more than any reported baseline (at most 4)—and the highest fused average in this comparison, numerically exceeding the strongest source model

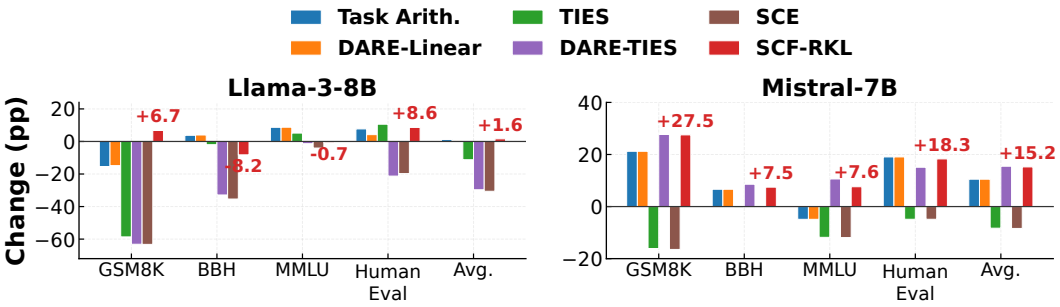


Figure 8: Performance change from the two-parent mean on four benchmarks and their equal-task average (percentage points). Left: Llama-3-8B fusion. Right: Mistral-7B fusion.

Table 2: Four-task means for the selected-pair recent-baseline extension (%) under the *reconciled canonical protocol*. Detailed scores are in Appendix B.8; the legacy primary tables use the original submitted protocol.

Pair	SCF-RKL	TSV	ISO-CTS	DELLA	Breadcrumbs
Mistral-7B	**43.26**	37.23	31.89	40.69	42.07
LLaMA3-8B	**52.39**	27.95	17.98	21.97	21.11

by 0.86 points (Table 23, Appendix). The corresponding paired-bootstrap interval includes zero (Table 10), so we describe this as numerically higher and statistically competitive rather than a significant parent-level improvement. The reported baselines suffer >97% generation dysregulation on GSM8K (Figure 1, left), collapsing scores by >68 points; SCF-RKL maintains <1% dysregulation with a 0.52-point decrease from the parent mean.

14B scale. SCF-RKL fuses TinyR1-Preview-Math-14B and TinyR1-Preview-Code-14B, attaining the highest fused score on 7 of 10 reported benchmarks and the highest fused average in this comparison (Table 24, Appendix). LiveCodeBench improves by $+6.55$ over the better parent (62.46 vs. 55.91). The corrected ten-task mean is 61.04, versus 60.96 for the Code parent, a margin of about 0.09 points before rounding the means. No significance claim is made for this corrected comparison (Table 10). On GSM8K, the evaluated baselines incur >99% dysregulation, dropping scores by >20 points; SCF-RKL maintains 0.08% dysregulation with a 0.18-point decrease from the parent mean.

7B/8B scale. SCF-RKL remains stable when fusing models with large capability gaps (source model average gap: 15 points; Tables 5 and 6, Appendix). On Mistral-7B, SCF-RKL improves over the source model average on all benchmarks, with an average gain of 15.2 points (Figure 8). On LLaMA-3-8B, where MAmmoTH2 itself exhibits near-100% dysregulation on GSM8K, the dense baselines in this comparison reach 100% dysregulation; SCF-RKL maintains only 0.38%, achieving a $+6.74$-point gain on GSM8K and a $+1.60$-point gain in overall average relative to the parent mean (Figure 8).

Additional recent-baseline stress tests. Table 2 compares TSV-Merge, ISO-CTS, DELLA, and Breadcrumbs on the two 7B/8B pairs. This selected-pair extension is not included in the primary 14B/32B comparison; per-task scores, adverse cases, configurations, and matched GSM8K stability tests are in Appendix B.8.

SCF-RKL has the highest mean on both pairs, but not on every task; we therefore claim a competitive capability–stability trade-off, not uniform dominance.

Safety + reasoning fusion. As shown in Figure 9, SCF-RKL fuses a safety-hardened model with a reasoning-specialized model, achieving consistent reasoning gains while **improving safety by ∼3 points** on average across 7 safety benchmarks—avoiding the typical safety–capability trade-off.

Vision-domain generalization. Four 2-in-1, two 4-in-1, and one final merge produce an 8-in-1 ViT-B/32 checkpoint. Including EuroSAT, SCF-RKL averages 60.14%, versus 58.25% for Task Arithmetic and 41.46% for SCE (Appendix Table 8).

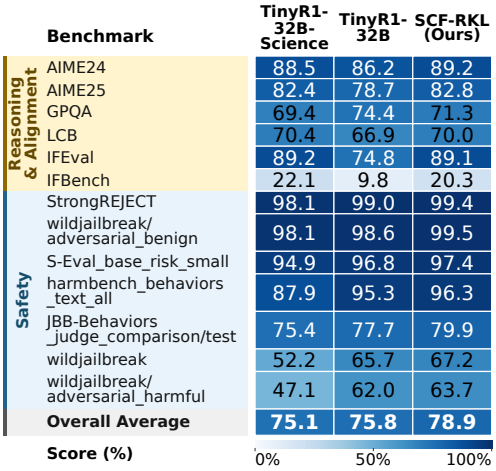


Figure 9: Comparison of two DeepSeek-R1-Distill-Qwen2.5-32B-derived models and the SCF-RKL fused model across 13 benchmarks.

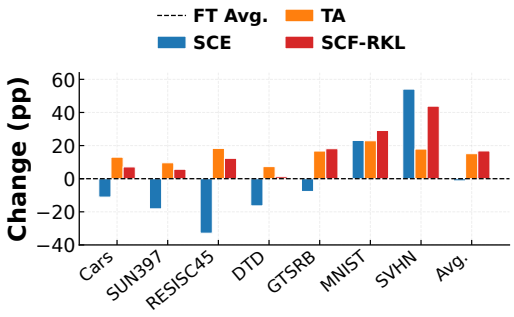


Figure 10: Vision accuracy improvement over FT Avg baseline across 7 image classification benchmarks.

6 Related work

Parameter-space merging. Task Arithmetic [1] adds task vectors; TIES [2] resolves sign conflicts; DARE [17] randomly drops entries; SCE [3] suppresses conflicting directions; and DELLA [19] uses magnitude-guided dropout. AdaMerging [38] and LED-Merging [4] use auxiliary data, whereas Breadcrumbs [20] constructs sparse delta masks and PCB-Merging [39] uses inter-model competition. These approaches differ in both the coordinates they modify and how they combine selected values. SCF-RKL isolates the selection step: its binary mask copies complementary parameters into an anchor, so each updated coordinate retains a value from one of the two sources. This design motivates comparing selection criteria and realized sparsity separately, and evaluating generation stability alongside task accuracy instead of inferring stability from a higher benchmark mean.

Recent structure-aware baselines. TSV-Merge reduces low-rank interference with task singular vectors [9]; ISO-CTS separates shared and task-specific subspaces [18]. Their selected-pair coverage, together with DELLA and Breadcrumbs, is reported in Appendix B.8.

KL divergence in model composition. Reverse KL is established in variational inference [8] and distillation [40]; SCF-RKL uses it as a parameter-level selection criterion once an anchor order is supplied. Here, the probability vectors are deterministic row-wise transformations of weights. Their divergence supplies an anchor-conditioned ranking signal for proxy discrepancies; its direction does not by itself establish causal importance for generation. The intermediate-logit correlation and criterion ablations provide empirical checks of this signal, while the proxy-space bounds remain distinct from guarantees about token-level outputs.

7 Conclusion and limitations

We document generation dysregulation as an observed merging failure and propose SCF-RKL, a sparse, anchor-ordered, parameter-only method. Across the reported 7B–32B evaluations it maintains L-GDR below 1% under the reference protocols and offers a competitive capability–stability trade-off; the proxy-space analysis is heuristic motivation rather than an end-to-end guarantee.

Limitations include the row-softmax proxy (including un-swept normalization and proxy-temperature choices), the non-exact realized densities in the criterion ablation, and the dependence of L-GDR on the stated length, decoding, stopping, and budget protocol. Baseline configurations are not an exhaustive tuning study, and the reported intervals quantify evaluation-item uncertainty but not decoding-seed variability. The LLM evaluation is pairwise and requires a supplied anchor order. The vision study demonstrates one balanced 8-in-1 tree, not order invariance or 14/16/24-model scaling. Automatic ordering, native many-way fusion, and evaluation on encoder–decoder and mixture-of-experts architectures remain open.

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Competing interests. The authors' institutional affiliations are listed on the first page. The authors declare no additional financial relationships outside the submitted work that could be perceived to influence this research.

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A Extended theoretical analysis

A.1 Per-matrix-type Spearman correlation: full numerical results

Table 3 provides the complete per-matrix-type Spearman ρ and p-values underlying Figure 3 in the main text. Two patterns emerge: (1) attention projections correlate more strongly than MLP projections (Mistral: 0.76 vs. 0.68; Qwen: 0.88 vs. 0.73), consistent with the attention mechanism's more direct role in sequential token prediction; (2) peak $\rho = 0.93$ (`q_proj`, Mistral-7B) demonstrates near-perfect rank correspondence for query projections, the most structurally sensitive component. Activation cosine distance gives consistent results (r=0.44–0.88; Appendix A.1), confirming the correlation is robust across metrics.

Table 3: Per-matrix-type Spearman ρ between param-level reverse KL and logit-level reverse KL (§3.3). All $p < 10^{-3}$; **bold**: largest ρ per model.

	Mistral-7B		**Qwen2.5-14B**	
Type	ρ	p	ρ	p
`q_proj`	**0.93**	9.6×10^{-15}	0.87	4.6×10^{-16}
`k_proj`	0.73	1.9×10^{-6}	**0.90**	3.1×10^{-18}
`v_proj`	0.70	6.8×10^{-6}	0.84	9.7×10^{-14}
`o_proj`	0.66	4.2×10^{-5}	**0.90**	4.4×10^{-18}
`gate_proj`	0.80	4.1×10^{-8}	0.75	1.2×10^{-9}
`up_proj`	0.59	3.4×10^{-4}	0.74	2.6×10^{-9}
`down_proj`	0.66	4.2×10^{-5}	0.70	3.5×10^{-8}
Attn avg.	*0.76*		*0.88*	
MLP avg.	*0.68*		*0.73*	

A.2 Second-order expansion and its scope

For completeness, we provide the second-order Taylor expansion of the reverse KL divergence around the anchor parameters. Consider a single row i (omitting subscript for clarity). Let $\theta_0 = W_{ai,:}$ and $\theta_1 = W_{ci,:}$, with $\Delta = \theta_1 - \theta_0$. The reverse KL divergence $D_{\mathrm{KL}}^{\mathrm{rev}}(q_0 \| q_1)$ where $q_k = \mathrm{softmax}(\theta_k)$ admits a second-order expansion:

$$D_{\mathrm{KL}}^{\mathrm{rev}}(q_0 \| q_1) \approx \frac{1}{2} \sum_j \Delta_j^2 \cdot \mathcal{F}_{jj} \tag{8}$$

where $\mathcal{F}_{jj}$ are the diagonal entries of the Fisher information matrix. Under the diagonal approximation, $\mathcal{F}_{jj} \approx q_{0,j}(1 - q_{0,j})$ captures the local curvature of the KL divergence with respect to parameter j.

This expansion is formally valid only for small $\|\Delta\|_2$; since our method selects coordinates with *large* $|\Delta_{i,j}|$ (the high-saliency tail), the approximation may not be tight for the selected subset.

From Fisher approximation to the SCF-RKL saliency. The coordinate-wise second-order approximation (Eq. 8) motivates selecting coordinates with large $\Delta_{i,j}^2 \cdot \mathcal{F}_{jj}$. We make two heuristic substitutions to obtain a row-weighted coordinate score:

(i) Per-coordinate Fisher → row-level summary. Computing $\mathcal{F}_{jj} = q_{0,j}(1 - q_{0,j})$ for every coordinate (i, j) costs $O(d_{\text{out}} \cdot d_{\text{in}})$ per matrix—the same as the full parameter count—but requires materialising the Fisher diagonal explicitly. We instead use the *row-level* $D_{\text{KL}}^{\text{rev}}(q_i \| p_i)$ as a summary of the displacement-dependent discrepancy within row i:

$$D_{\text{KL}}^{\text{rev}}(q_i \| p_i) = \sum_j q_{i,j} \log \frac{q_{i,j}}{p_{i,j}} \approx \frac{1}{2} \sum_j q_{i,j}(1 - q_{i,j}) \cdot \Delta_{i,j}^2 \tag{9}$$

Both the Fisher diagonal and row-level divergence can be computed in $O(d_{\text{in}})$ time per row. The latter is a design choice for summarizing row discrepancies, not an asymptotic improvement over the Fisher diagonal. Large divergence can reflect perturbation magnitude as well as curvature.

(ii) Squared magnitude → absolute magnitude. We use $|\Delta_{i,j}|$ instead of $\Delta_{i,j}^2$ (i.e., the ℓ_1 rather than ℓ_2 measure of deviation). This substitution does not change the intra-row ranking (both are monotone in $|\Delta_{i,j}|$) but improves robustness to outlier entries that would disproportionately inflate $\Delta_{i,j}^2$ under heavy-tailed weight distributions.

Combining both substitutions yields the SCF-RKL saliency score:

$$\mathcal{I}_{i,j} = |\Delta_{i,j}| \cdot D_{\text{KL}}^{\text{rev}}(q_i \| p_i).$$

We therefore use this analysis as a *motivating heuristic* to identify which coordinates are most sensitive to perturbation, rather than as a quantitative bound. The resulting saliency criterion is validated empirically (§3.4) rather than claimed to be second-order optimal.

A.3 Intra-row vs. inter-row selection

We emphasize the *two distinct roles* of the saliency components:

- **Intra-row ranking** is determined by $|\Delta_{i,j}|$: within a fixed row i, the reverse-KL term $D_{\text{KL}}^{\text{rev}}(q_i \| p_i)$ is a constant that does not affect the ordering of coordinates. Thus, *within each row*, the selection reduces to magnitude-based ranking.
- **Inter-row prioritization** is governed by $D_{\text{KL}}^{\text{rev}}(q_i \| p_i)$: rows with higher proxy divergence receive proportionally higher saliency scores, causing their coordinates to be prioritized in the *global* threshold selection (across all rows of a weight matrix).

This two-level mechanism is the key distinction from pure magnitude selection: magnitude treats all rows equally, while reverse-KL saliency concentrates the fusion budget on rows where the complementary model offers the most structurally distinct information. The ablation in §3.4 reports a 1.7-point aggregate advantage over magnitude-only selection (43.19 vs. 41.52).

A.4 Relationship to exact second-order optimality

Under the diagonal Fisher approximation (Eq. 8), the second-order-optimal mask would select coordinates with the largest $\Delta_{i,j}^2 \cdot \mathcal{F}_{i,jj}$. Our saliency $\mathcal{I}_{i,j} = |\Delta_{i,j}| \cdot D_{\text{KL}}^{\text{rev}}(q_i \| p_i)$ introduces two approximations: (i) replacing per-coordinate Fisher entries with a row-level summary, and (ii) using $|\Delta_{i,j}|$ instead of $\Delta_{i,j}^2$ for robustness to outlier magnitudes. For a fixed row-level weight, replacing squared magnitude by absolute magnitude preserves coordinate ordering. Replacing the coordinate-specific Fisher entries by a row-level scalar need not preserve their ranking. In §3.4, reverse-KL saliency is empirically useful among the alternative criteria evaluated in Figure 4.

A.5 Proof of entropy preservation bound

Proof of Proposition 3.1. Let $p = \mathrm{softmax}(z)$ and $g(z) = H(p)$, using natural logarithms. Differentiation gives

$$\frac{\partial g}{\partial z_j} = p_j \left(\sum_k p_k z_k - z_j \right).$$

For $z \in [-C, C]^{d_{\text{in}}}$, each component has magnitude at most $2Cp_j \leq 2C$. Integrating along the line segment between the anchor and fused rows therefore yields

$$|H(q_{f,i}) - H(q_i)| \leq 2C \|M_i \odot \Delta_i\|_1 = 2C \sum_{j \in \mathcal{S}_i} |\Delta_{i,j}|.$$

Rearranging gives the stated lower bound with $L = 2C$. □

A.6 On the Lipschitz constant

The constant $L = 2C$ is a sufficient bound on the specified box, not a claim of tightness. Although $\|M_i \odot \Delta_i\|_1 \leq \|\Delta_i\|_1$, a small replacement fraction alone does not ensure a small perturbation: selected coordinates can have large magnitudes. Whether the entropy bound is informative depends on the actual row values and selected updates.

A.7 Proof of subspace rotation bound

Proof of Theorem 3.2. Write $e = \|E\|_2$. Weyl's inequality implies $\sigma_r(W_a + E) \geq \sigma_r(W_a) - e$, so its separation from the trailing singular values of W_a is at least $\delta - e > 0$. Also, the perturbed gap is at least $\delta - 2e > 0$. Let $\tilde{U}, \tilde{V}, \tilde{\Sigma}$ denote the leading perturbed singular triplets. The two Wedin residuals are $R = W_a \tilde{V} - \tilde{U}\tilde{\Sigma} = -E\tilde{V}$ and $S = W_a^\top \tilde{U} - \tilde{V}\tilde{\Sigma} = -E^\top \tilde{U}$, so $\|R\|_2, \|S\|_2 \leq e$. The spectral-norm residual form of Wedin's theorem bounds the sine of the largest principal angle by $\sqrt{\|R\|_2^2 + \|S\|_2^2}/(\delta - e)$, which gives the result. □

A.8 Validity of the spectral gap condition

The original geometric audit reports $\|E\|_2/\delta < 0.3$ for the measured 7B matrices at $\alpha = 1.5$. This satisfies the sufficient condition $\|E\|_2 < \delta/2$. Neither a low replacement fraction nor a visually decaying singular spectrum alone establishes the condition; it requires measuring the perturbation norm and the gap at the chosen rank r.

A.9 Evidence analysis: expanded discussion

Criterion comparison: The 1.7-point gap between RKL and magnitude supports the usefulness of inter-row weighting in this ablation. Both RKL (43.19) and magnitude (41.52) use magnitude for *intra-row* ranking (Appendix A.3). The difference is that RKL additionally weights rows by $D_{\text{KL}}^{\text{rev}}(q_i \| p_i)$ in the *inter-row* global selection. The observed advantage of RKL—including generation-heavy tasks (GSM8K: +3.03, HumanEval: +1.68)—is consistent with useful inter-row weighting, subject to the unequal-density limitation. On MMLU, a multiple-choice knowledge-recall task with minimal generation demands, magnitude performs comparably (36.38 vs. 38.78), consistent with the entropy-preservation analysis (§3.2): distributional awareness matters most when generative coherence is required.

Threshold sensitivity: $\alpha = 1.5$ has the highest observed average. See Table 4 (§4) for the full sweep. The anchor model (MathCoder2-Mistral-7B, avg. 20.23) excels at BBH but is weak on GSM8K and HumanEval; the complementary model (Mistral-7B-Instruct-v0.2, avg. 36.74) is stronger overall but neither model alone exceeds 37 on average. As α increases, the fusion becomes sparser—fewer complementary parameters are incorporated—and the fused model gradually reverts toward the anchor model. This is clearly visible at $\alpha \geq 3.0$: the fused model's average (≈20–21) converges to the anchor model's standalone score (20.23), confirming that high α effectively recovers the unmodified anchor. Performance is non-monotonic over $\alpha \in [1.0, 2.0]$, peaking at $\alpha = 1.5$; the observed range in this Mistral ablation is 4.22 points (38.97–43.19). The fused model is above both parent averages throughout this interval (best parent: 36.74), but this is a sensitivity result

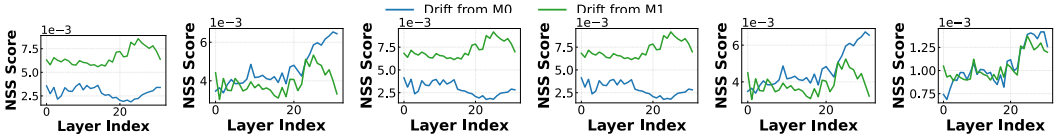
rather than an invariance claim or a guarantee outside the evaluated pair. Two failure modes bound this optimum: (i) *over-fusion* ($\alpha = 0$, avg. 31.17): 50% of parameters are replaced, introducing excessive low-saliency noise that disrupts coherence (HumanEval drops to 7.62); (ii) *under-fusion* ($\alpha \geq 2.5$, avg. $<$27): too few complementary parameters pass the threshold, and the model reverts to anchor-model behavior. The best observed average occurs at the pre-specified $\alpha = 1.5$ setting (Table 4).

A.10 IQR scale ablation: per-benchmark results

Table 4: Ablation on IQR scale α (reverse-KL saliency). Anchor: MathCoder2-Mistral-7B; Compl.: Mistral-7B-Instruct-v0.2. $\tau{=}\mu{+}\alpha(\text{Q3}{-}\text{Q1})$; **bold**: best per col.

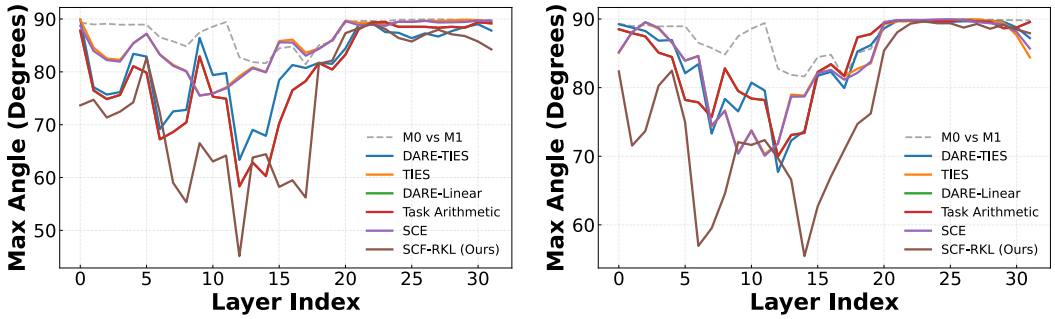
α	Regime	BBH	GSM8K	HEval	MMLU	Avg
Anchor		46.66	14.95	0.15	19.14	20.23
Compl.		47.38	46.76	9.60	43.23	36.74
0	∼50%	39.22	33.74	7.62	**44.10**	31.17
1.0	Moderate	50.81	52.58	15.24	38.54	39.29
1.5	**Default**	54.29	**57.26**	**22.41**	38.78	**43.19**
2.0	Conserv.	**56.34**	45.20	20.58	33.76	38.97
2.5	Aggress.	46.78	24.51	6.71	28.19	26.55
3.0	V. sparse	40.02	21.40	1.22	19.64	20.57
3.5	V. sparse	44.70	17.93	0.30	17.56	20.12
4.0	Extr. sp.	47.16	17.19	0.76	19.21	21.08

A.11 Geometric analysis: full-layer and NSS results



(a) DARE-TIES (b) TIES (c) DARE-Linear (d) Task Arith. (e) SCE (f) SCF-RKL

Figure 11: Normalized Spectral Shift (NSS) for all fusion methods. Lower NSS values indicate better preservation of the parent models' spectral properties. Note: The y-axis scale for SCF-RKL has been adjusted for better visualization.



(a) Max angle deviation vs M0 (b) Max angle deviation vs M1

Figure 12: Maximum principal angle across all layers for different fusion methods. Lower values indicate better geometric consistency with the parent models.

B Additional experimental results

B.1 Accuracy change by blocking layers

Figure 13 presents a layer-wise *blocking* diagnostic on three safety benchmarks (S-Eval, StrongREJECT, and WildJailbreak), each evaluated with both the overall metric and the S3 subset. Concretely, starting from the merged model M_f, we *restore* the parameters of selected transformer layers to the base model M_0 (i.e., replace the fused layer with the corresponding layer from M_0) while keeping all other layers unchanged, and then measure the score change relative to the original merged model. A negative score change indicates that blocking (restoring to M_0) hurts performance, implying that the fused parameters in that layer contribute positively to the merged model's safety behavior.

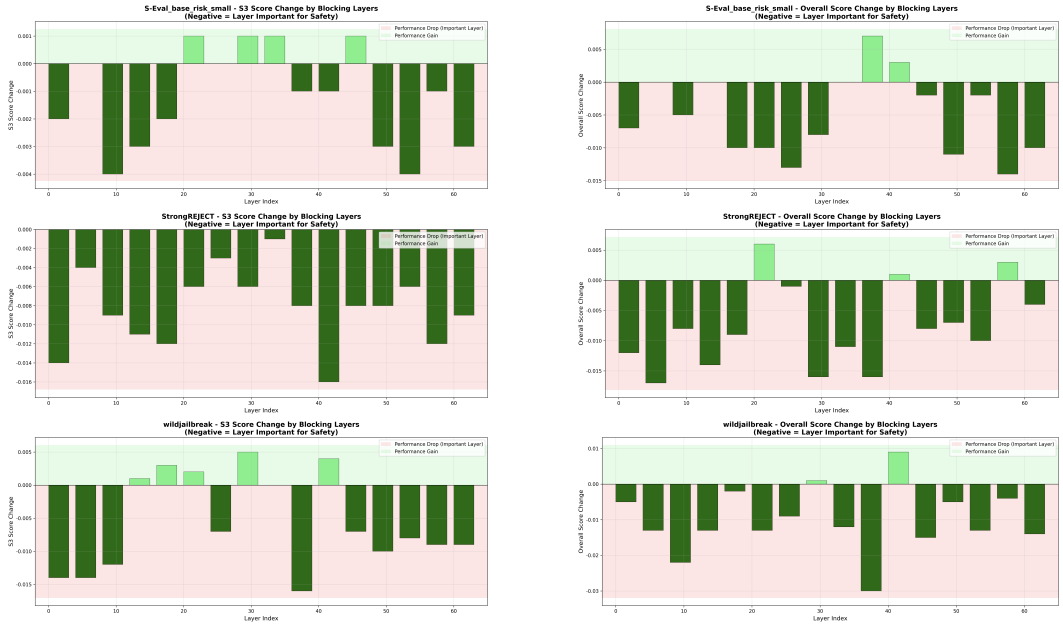


Figure 13: Score changes after restoring individual layers to the base model on three safety benchmarks.

Across all three benchmarks, most blocked layers lead to a score drop, and this trend holds for both S3 and overall metrics. This supports the interpretation that the safety improvements are distributed across multiple layers rather than driven by one isolated component. The magnitude is non-uniform across depth, and a few layers show small positive changes, consistent with noisy or task-irrelevant updates. Overall, Figure 13 provides a layer-localized diagnostic, but does not by itself rule out evaluation variance.

B.2 7B/8B fusion: per-benchmark accuracy

On Mistral-7B, SCF-RKL achieves an average of 43.70—comparable to the best baseline (DARE-TIES: 43.97)—while outperforming all others by a large margin, particularly on GSM8K (+6.30 over Task Arithmetic) and HumanEval (+23.17 over TIES). The near-tie with DARE-TIES on average conceals a qualitative difference: DARE-TIES achieves its score by incurring high dysregulation on other tasks, whereas SCF-RKL maintains near-zero dysregulation throughout. On LLaMA-3-8B, the fusion scenario is harder because MAmmoTH2 itself exhibits near-100% L-GDR on GSM8K. Under the original submitted protocol, SCF-RKL achieves the highest fused average (53.01) and preserves GSM8K at 71.68; the reconciled canonical protocol gives 52.39 and 71.97, respectively (Table 15).

B.3 Vision accuracy details

The per-task rows in Table 7 are the seven task-level results available in the current evaluation archive. The EuroSAT aggregate is included for completeness, but its task-level prediction artifact is not

Table 5: Meta-Llama-3-8B-Instruct (base0) and MAmmoTH2-8B-Plus (base1). All metrics are pass@1 under the *original submitted protocol*; Avg. is the equal mean of the displayed task scores. Best overall is underlined; best fused is **bolded**.

Methods	GSM8K↑	BBH↑	MMLU↑	HumanEval↑	Avg.
base0	73.35	65.78	55.11	43.14	59.35
base1	56.54	53.92	38.68	24.70	43.46
Task Arith.	49.56	63.63	55.58	41.62	52.60
DARE-Linear	50.11	**63.82**	**55.64**	38.11	51.92
TIES	6.31	57.89	52.04	**44.51**	40.19
DARE-TIES	1.80	26.98	45.54	12.65	21.74
SCE	1.72	24.48	42.85	14.18	20.81
SCF-RKL (Ours)	**71.68**	51.64	46.17	42.53	**53.01**

Table 6: Mistral-7B-Instruct-v0.2 (base0) and MathCoder2-Mistral-7B (base1). All metrics are pass@1 under the *original submitted protocol*; Avg. is the equal mean of the displayed task scores. Best overall is underlined; best fused is **bolded**.

Methods	GSM8K↑	BBH↑	MMLU↑	HumanEval↑	Avg.
base0	46.76	47.38	43.23	9.60	36.74
base1	14.95	46.66	19.14	0.15	20.23
Task Arith.	52.08	53.66	26.29	**23.93**	38.99
DARE-Linear	52.08	53.66	26.29	**23.93**	38.99
TIES	14.75	46.60	19.37	0.00	20.18
SCE	14.39	46.67	19.28	0.00	20.09
DARE-TIES	**58.53**	**55.59**	**41.78**	19.97	**43.97**
SCF-RKL (Ours)	58.38	54.47	38.79	23.17	43.70

Table 7: Vision accuracy (%). SCF-RKL surpasses fine-tuned average on all 7 benchmarks.

Method	Cars	SUN397	RESISC45	DTD	GTSRB	MNIST	SVHN	Avg
FT Avg	49.0	54.1	48.1	39.7	35.8	60.5	43.1	47.2
SCE	37.9	35.9	15.2	23.3	28.1	83.6	**97.2**	45.89
TA	**62.0**	**63.7**	**66.4**	**47.0**	52.5	83.4	61.0	62.29
SCF-RKL	56.1	59.8	60.4	40.9	**53.9**	**89.6**	86.9	**63.94**

Table 8: Eight-task macro average (%) for the recursively constructed 8-in-1 ViT-B/32 checkpoint, including EuroSAT. The balanced construction is four 2-in-1 merges, two 4-in-1 merges, and one 8-in-1 merge.

Method	SCF-RKL	Task Arithmetic	SCE
Eight-task macro average	**60.14**	58.25	41.46

currently available; we therefore do not reconstruct a EuroSAT cell from rounded averages. The exact leaf pairing and intermediate checkpoint identifiers are not retained in the available archive.

B.4 Metric protocol and complementary stability measures

The following definitions are operational measures of the fixed generation protocol used in our evaluation; max-hit and normal-stop are stopping-status rates rather than separate learned metrics. They are defined here as protocol-level diagnostics (rather than borrowed benchmark scores), so the definitions—not an external citation—determine their meaning and ensure that they can be reproduced from the returned completions.

The canonical parent reference uses the EvalScope templates shared by all methods; GSM8K uses the four-shot chain-of-thought prompt, with no method-specific stop sequence. Parent reference completions use $T = 0.6$, $n = 4$, and a 4,096-token completion budget. Robustness runs use matched temperature, budget, and presence-penalty settings listed in Table 18. EOS and the configured service stop are treated as normal stops. A service/request failure is tracked separately from a returned completion: it is excluded from the parent reference-eligible set and is not counted as a max-hit

Table 9: Generation-stability metrics used alongside L-GDR. All lengths use completion tokens and exclude prompts.

Metric	Definition
L-GDR@$k\times$	$P(L > km)$, $k \in \{5, 10, 20\}$, over all returned completions
Max-hit	$P(\texttt{finish_reason=length})$ or equivalent budget hit
Normal-stop	$P(\texttt{finish_reason=stop})$ via EOS/service stop
Rep-4	Mean $1 - U_4/A_4$, where U_4 and A_4 are distinct/all 4-grams
Hit accuracy	Accuracy restricted to completions with $L > km$
Semantic audit	Manual inspection of sampled length outliers for coherence

or an L-GDR event. The available archive does not retain failure counts or the exact template and service-stop configuration for every benchmark; these reporting gaps limit reproducibility.

The semantic audit is qualitative: sampled L-GDR outliers are checked for task relevance, coherent reasoning, and answer structure. It is used as a diagnostic alongside hit accuracy and Rep-4, not as a separately estimated population metric or a correctness guarantee.

Rep-4 is set to zero when fewer than four completion tokens are available. Request failures are excluded from the parent reference-eligible set and are tracked separately from returned-completion metrics; successful returned completions, including length-hit outputs, remain in the L-GDR denominator. The paired intervals below use question-cluster paired percentile bootstrap resampling (20,000 replicates for the L-GDR robustness tables; 2.5th and 97.5th percentiles) and describe finite evaluation-item uncertainty, not decoding-seed variability. For BBH/MMLU, resampling is stratified within each subset before taking the same equal-subset macro-average used for the point estimate.

Table 10: Strongest-parent comparisons. The 14B point estimate is recomputed from the ten displayed task scores; its earlier interval is not retained because it used a different aggregate and comparator. The 32B interval is from the archived paired-bootstrap audit (50,000 stratified replicates).

Scale	Comparator	SCF-RKL − parent [95% CI]
14B Math+Code	Code parent	$+0.09$ [unavailable]
32B Math+Science	Math parent	$+0.86$ $[-1.24,\ 2.88]$

The 14B mean scores are 61.042 for SCF-RKL, 60.956 for Code, and 60.035 for Math when computed from the displayed task scores. Thus, Code is the strongest parent under the equal-task average. The previously reported $+0.80\,[-0.27, 1.88]$ comparison against Math cannot be transferred to this corrected comparison. Matched prediction artifacts are unavailable, so we cannot recompute its interval and make no significance claim.

The 32B interval resamples matched evaluation items with identical indices for SCF-RKL and the fixed Math parent. BBH/MMLU resampling is stratified by subset, and each replicate retains the benchmark aggregation and equal-task average. The interval quantifies evaluation-item uncertainty, not decoding-seed variability; it includes zero. Its archived value is retained without claiming that the raw bootstrap has been independently reproduced.

B.5 Criterion density and alternative row criteria

The criterion ablation uses the same tensor-wise IQR construction and the same threshold multiplier, but it does not force an equal number of selected coordinates. We retain this common threshold protocol to compare the criteria as implemented; enforcing equal density would require criterion-specific thresholds and is a separate experiment. The realized densities below are therefore reported explicitly; they should not be read as an exactly matched-sparsity experiment.

On the common four-task 32B subset, RKL has the highest aggregate among the three evaluated divergence criteria, while FKL is slightly higher than JS. The complete ten-task comparison is available for RKL and FKL. These results support the statement that RKL is empirically useful among the tested criteria; they do not establish uniqueness, universal optimality, or an exactly matched-density advantage; an exactly density-matched criterion study remains outside this evaluation.

Table 11: Realized density audit and alternative row-level criteria. Density is global parameter-weighted selection on the audited Mistral-7B pair. The 32B column is the common four-task subset average from the criterion audit.

Criterion	7B density (%)	32B common-four Avg.	32B complete Avg.
Reverse KL (RKL)	5.4261	67.80	65.3
Magnitude	5.9844	–	–
Forward KL (FKL)	8.3863	67.08	64.5
JS divergence	7.6132	66.43	–

B.6 Effective baseline configurations and DARE sensitivity

The following audit records the values consumed by the checked implementations in the fixed comparisons. They characterize the reported runs and are not claims that every baseline is at a validation-tuned optimum.

Table 12: Effective baseline settings in the fixed comparisons. A density of one disables pruning in the checked implementation; consequently, configured rescaling is inactive in those runs.

Method	λ	Density	Normalize/rescale	Effective note
Task Arithmetic	1.0	1.0	no/no	fixed task-vector merge
TIES	1.0	1.0	yes/–	trimming inactive
DARE-Linear	1.0	1.0	no/no†	dropout inactive
DARE-TIES	1.0	1.0	no/no†	dropout inactive
SCE	–	–	–	select_topk=1

† The implementation accepts a rescale option, but `density=1` returns the tensor before pruning/rescaling. In the legacy Mistral-7B task-vector runs, one source also served as the base reference and was excluded from the active task-vector list; the LLaMA3-8B and TinyR1 runs used a separate common base and two active source vectors.

For the baseline-favorable DARE sensitivity audit, we evaluated the pre-specified grid $d \in \{0.1, 0.5, 1.0\}$ and $\lambda \in \{0.5, 1.0\}$ (six configurations per method), with all other settings fixed. The best GSM8K scores were 2.96 for DARE-Linear at $(d, \lambda) = (0.1, 0.5)$ and 2.73 for DARE-TIES at $(0.1, 1.0)$. The best HumanEval score for each was 25.61 (at $(0.5, 0.5)$ for DARE-Linear and $(1.0, 0.5)$ for DARE-TIES). All twelve grid points had 100% L-GDR@$10\times$ on GSM8K and HumanEval. Because the best configuration is selected after observing test scores, these maxima are reported as a best-in-grid sensitivity analysis (a test-set oracle), not as validation tuning or an exhaustive fair search. The per-configuration score files are unavailable; only the recorded maxima and grid-wide L-GDR summary can be reported.

B.7 Anchor direction across source pairs

Table 13 reports the direction-swap audit. These archived values use the parent ranking and averages recorded in the original direction audit. W/T/L counts benchmarks on which its originally designated lower-average anchor is better/tied/worse than the reverse direction. In particular, the corrected 14B table ranks Code above Math; this does not change which checkpoint was used in the archived runs. The direction-audit averages cannot be recomputed without their task-level outputs and are not used as corrected primary performance estimates.

Table 13: Anchor-direction audit under the *original submitted protocol* (original submitted aggregation).

Pair	Archived gap	W/T/L	Original lower-anchor	Original higher-anchor	Δ
Mistral-7B	16.5	4/0/0	43.7	37.5	+6.2
LLaMA3-8B	15.9	4/0/0	53.0	36.9	+16.1
TinyR1-14B	1.0	5/1/4	61.4	58.7	+2.7
TinyR1-32B	1.0	7/0/3	65.3	64.5	+0.8

The large-gap pairs favor one direction consistently in this audit, whereas the near-tied pairs show mixed task-level signs. These observations support evaluating direction sensitivity; they do not establish a universal lower-average-anchor rule or tree-order invariance.

B.8 Recent baseline scope and coverage

The four recent methods are included as protocol-matched stress tests on the Mistral-7B and LLaMA3-8B pairs, rather than as reruns of every primary benchmark at every model scale. The table lists the configurations with complete four-task capability results; generation-only stress results are reported separately. These rows are an explicitly scoped extension and do not alter the primary 14B/32B comparison.

Table 14: Coverage of additional recent baselines. The primary SCF-RKL comparison remains the broader benchmark suite in the main paper.

Source pair	Method	Capability tasks	GSM8K stability	Status
Mistral-7B	TSV-Merge	GSM8K/BBH/MMLU/HumanEval	complete	complete
Mistral-7B	ISO-CTS	GSM8K/BBH/MMLU/HumanEval	complete	complete
Mistral-7B	DELLA	GSM8K/BBH/MMLU/HumanEval	complete	complete
Mistral-7B	Breadcrumbs	GSM8K/BBH/MMLU/HumanEval	complete	complete
LLaMA3-8B	TSV-Merge	GSM8K/BBH/MMLU/HumanEval	complete	complete
LLaMA3-8B	ISO-CTS	GSM8K/BBH/MMLU/HumanEval	complete	complete
LLaMA3-8B	Breadcrumbs	GSM8K/BBH/MMLU/HumanEval	complete	complete
LLaMA3-8B	DELLA	GSM8K/BBH/MMLU/HumanEval	complete	complete

Table 15: Canonical four-task capability scores for the recent-baseline stress tests under the *reconciled canonical protocol*. Scores use the reconciled canonical per-question aggregation; the legacy primary tables retain the original submitted protocol.

Pair	Method	GSM8K	BBH	MMLU	HumanEval	Avg.
Mistral-7B	SCF-RKL	57.26	54.36	39.00	22.41	43.26
	TSV-Merge	47.37	43.12	29.63	28.81	37.23
	ISO-CTS	28.96	42.54	26.94	29.12	31.89
	DELLA	55.36	49.41	38.50	19.51	40.69
	Breadcrumbs	56.10	53.95	41.77	16.46	42.07
LLaMA3-8B	SCF-RKL	71.97	51.47	46.16	39.94	52.39
	TSV-Merge	0.51	33.65	37.69	39.94	27.95
	ISO-CTS	1.72	22.77	36.61	10.82	17.98
	Breadcrumbs	1.54	24.95	42.70	15.24	21.11
	DELLA	1.21	23.67	47.29	15.70	21.97

These additional-baseline rows use the reconciled per-question aggregation: BBH and MMLU are subset-macro averages, and the four-task Avg. is an equal mean of GSM8K, BBH, MMLU, and HumanEval. The primary tables retain the original task-level scores, with their Avg. columns recomputed as equal-task means of the displayed values. Consequently, values such as 43.70 versus 43.26 (Mistral) and 53.01 versus 52.39 (LLaMA3) should not be interpreted as method changes or new model checkpoints: they reflect the two reporting protocols and their associated archived evaluation snapshots. The available archive does not identify the exact contribution of aggregation and decoding changes to each difference. We therefore keep the protocols separate and do not attribute residual discrepancies to sampling noise. All recent-baseline comparisons use the reconciled canonical protocol.

The recent-baseline paired percentile-bootstrap intervals retain adverse task outcomes. Matched questions are resampled, with stratification within BBH/MMLU subsets and the same equal-task average. These intervals quantify finite-item uncertainty, not decoding-seed variability. The archived capability summaries do not retain the replicate count; it should not be inferred from the separate 20,000-replicate L-GDR audit.

Table 16: Mistral-7B score differences, SCF-RKL minus comparator [paired 95% CI], in percentage points, under the canonical protocol.

Benchmark	vs. TSV-Merge	vs. ISO-CTS
GSM8K	$+9.89$ $[8.00, 11.77]$	$+28.30$ $[26.40, 30.16]$
BBH	$+11.24$ $[10.27, 12.22]$	$+11.82$ $[10.92, 12.72]$
MMLU	$+9.38$ $[8.38, 10.35]$	$+12.06$ $[11.11, 13.00]$
HumanEval	-6.40 $[-11.28, -1.52]$	-6.71 $[-11.43, -2.13]$
Four-task mean	$+6.03$ $[4.65, 7.36]$	$+11.37$ $[10.08, 12.63]$

Table 17: LLaMA3-8B four-task mean differences, SCF-RKL minus comparator [paired 95% CI], in percentage points, under the canonical protocol.

Comparator	Mean difference [95% CI]
TSV-Merge	$+24.44\ [23.22, 25.65]$
ISO-CTS	$+34.40\ [32.97, 35.90]$
DELLA	$+30.42\ [28.93, 31.91]$
Model Breadcrumbs	$+31.28\ [29.84, 32.74]$

Table 18: Cross-pair GSM8K stress results at both matched decoding settings. Each cell is score/L-GDR@10× (%). These rows are kept separate from the canonical four-task capability scores.

Pair/setting	SCF-RKL	TSV	ISO-CTS	DELLA	Breadcrumbs
Mistral-7B, $T = 0$	60.65/1.29	51.63/6.37	31.54/44.58	59.89/.61	58.68/.68
Mistral-7B, $T = 1$	46.55/.61	40.03/9.63	27.82/3.56	50.80/–[†]	52.39/.45
LLaMA3-8B, $T = 0$	77.10/.46	22.90/0	2.50/100	1.90/100	2.50/100
LLaMA3-8B, $T = 1$	62.17/.38	41.24/0	1.59/99.92	1.44/100	1.14/100

[†] The archived summaries disagree on DELLA's Mistral $T = 1$ L-GDR (0.61% versus 0.68%). Without the original completions, we leave that cell unresolved rather than choosing one value. Both summaries place it below 1%, but they do not establish an exact ordering against SCF-RKL's 0.61%.

Effective configurations. The recent-baseline runs use fixed implementation settings rather than pair-specific validation: TSV-Merge uses its recorded $\alpha = 1$ FP32 merge; ISO-CTS uses $\alpha = 1$ and common-space fraction 0.8; DELLA uses $\lambda = 0.5$, density 0.5, $\epsilon = 0.05$, and seed 0; and Model Breadcrumbs uses density 0.9 and $\gamma = 0.01$. These settings are disclosed to make the stress tests reproducible, not to claim that any baseline is at a universal optimum. The primary SCF-RKL tables remain the broad comparison; the recent-baseline rows are an explicitly labeled, non-exhaustive extension. For the GSM8K stress table, all methods share $n = 1$, a 16,384-token budget, and presence penalty 1; the two settings use $T = 0$ with $(\texttt{top_p}, \texttt{top_k}) = (1, -1)$ and $T = 1$ with $(0.95, 20)$, respectively. The parent-derived L-GDR median and tokenizer are frozen across methods within a row.

B.9 Generation dysregulation rates: ablation details

The Mistral-7B GSM8K reference median is fixed at $m = 159$ completion tokens, giving a 1,590-token threshold for L-GDR@10×. For the TSV temperature sweep in Table 1, the SCF-RKL accuracy gains and paired 95% intervals are $+9.02\ [5.99, 12.05]$ at $T = 0$, $+9.61\ [7.70, 11.50]$ at $T = 0.6$, and $+6.52\ [3.18, 9.86]$ at $T = 1$.

Table 19: Matched Mistral-7B GSM8K budget and presence-penalty (PP) sweep. Stability cells are SCF-RKL/ISO-CTS; L-GDR and max-hit are percentages. Score differences are in percentage points.

Budget/PP	Score gain [95% CI]	L-GDR@10×	Max-hit	Rep-4
4096/1	$+29.11\ [26.00, 32.30]$	1.14/45.64	1.14/45.64	.148/.408
8192/1	$+29.19\ [26.08, 32.30]$	1.14/45.19	.91/44.88	.149/.448
16384/1	$+29.64\ [26.46, 32.83]$	1.29/45.79	0/45.72	.148/.474
8192/0	$+60.50\ [57.77, 63.15]$	4.62/100	4.55/100	.198/.962

ISO-CTS's L-GDR@10× hits in the reported sweep have only 0–3.85% accuracy and elevated repetition. In the matched LLaMA3-8B $T = 1$ run, ISO-CTS has 99.92% L-GDR despite 100% normal stops and 0% max-hit. Inspected completions include thousands of task-irrelevant tokens before EOS. These diagnostics distinguish abnormal length from budget exhaustion and show why a low max-hit rate alone is insufficient.

The intervals use question-cluster paired bootstrap resampling; they quantify finite-evaluation uncertainty and do not represent decoding-seed variability. At presence penalty 1, the corresponding max-hit gaps are -44.50, -43.97, and -45.72 percentage points for 4096, 8192, and 16384 tokens, respectively, with all reported intervals excluding zero. Absolute L-GDR values remain protocol-dependent, so these tables are interpreted together with accuracy, normal-stop, Rep-4, hit accuracy, and semantic audit.

Table 20: Paired bootstrap 95% intervals for SCF-RKL minus TSV L-GDR differences on matched Mistral-7B GSM8K runs (percentage points). Negative values favor SCF-RKL.

T	Δ@5×	Δ@10×	Δ@20×
0.0	$-5.23\,[-6.75, -3.71]$	$-5.08\,[-6.60, -3.64]$	$-5.00\,[-6.44, -3.56]$
0.6	$-12.76\,[-13.93, -11.60]$	$-12.34\,[-13.42, -11.30]$	$-11.77\,[-12.77, -10.78]$
1.0	$-13.72\,[-16.00, -11.52]$	$-9.02\,[-10.69, -7.43]$	$-4.55\,[-5.76, -3.41]$

Table 21: Matched SCF-RKL minus ISO-CTS L-GDR differences under the budget and presence-penalty sweep (percentage points).

Budget/PP	Δ@5×	Δ@10×	Δ@20×
4096/1	$-44.50\,[-47.23, -41.77]$	$-44.50\,[-47.23, -41.77]$	$-44.50\,[-47.23, -41.70]$
8192/1	$-44.05\,[-46.78, -41.32]$	$-44.05\,[-46.78, -41.32]$	$-43.97\,[-46.70, -41.24]$
16384/1	$-44.43\,[-47.23, -41.70]$	$-44.50\,[-47.23, -41.77]$	$-44.50\,[-47.31, -41.77]$
8192/0	$-95.30\,[-96.44, -94.16]$	$-95.38\,[-96.51, -94.16]$	$-95.45\,[-96.51, -94.31]$

Table 22 reports generation dysregulation rates for the controlled ablations in §3.4, measured on Mistral-7B fusion on GSM8K.

Ablation 1: selection criterion under a shared IQR protocol. All four criteria use α=1.5 and the same tensor-wise IQR construction, but their realized densities differ (RKL 5.4261%, magnitude 5.9844%, forward KL 8.3863%, and JS 7.6132% in the audited 7B run; Table 11). This is therefore not an exactly matched-sparsity comparison; it evaluates the criteria as used under the common thresholding protocol. Random selection produces 72.3% L-GDR, showing that binary masking alone, despite eliminating phantom parameters (Proposition 3.3), is insufficient in this evaluated ablation. Sign disagreement (4.7%) improves over random but does not include a magnitude-based coordinate ranking. Magnitude selection (1.0%) captures parameter-level deviation but ignores inter-row structural divergence. RKL (0.5%) has the lowest rate in this comparison and couples coordinate-level magnitude with row-level anchor-centric divergence—the two-level score described in §3.1.

Ablation 2: sparsity level (fixed RKL criterion). Fixing the RKL criterion and varying α reveals a non-monotonic pattern. At $\alpha = 0$, the median threshold replaces approximately half the parameters and L-GDR is 4.4%. The lowest observed rate occurs at α=1.5 (fixed IQR setting, 0.5%), where IQR-bounded sparsity restricts updates to the statistical tail of the saliency distribution—parameters with both large deviation and high structural divergence. Beyond $\alpha \geq 2.5$, over-sparsity progressively strips complement contributions, and the fused model monotonically recovers the anchor's own dysregulation profile: 34.9% (α=2.5) → 53.6% (α=3.0) → 69.4% (α=3.5) → 69.5% (α=4.0), asymptotically approaching the anchor's 73.4%. This monotonic pattern is consistent with over-sparse fusion approaching the anchor model and inheriting its observed failure modes; it is not an end-to-end causal confirmation.

Joint interpretation. The two ablations provide complementary empirical evidence: Ablation 1 compares selection criteria under the shared IQR protocol (0.5% vs. 72.3%), while Ablation 2 varies the sparsity setting with RKL fixed (0.5% vs. 69.5%). The lowest observed rate in this ablation occurs with RKL-structured selection and IQR-bounded sparsity, the two core design choices of SCF-RKL; this is an empirical comparison rather than a universal requirement. These measurements connect the three design choices identified in §2 (binary selection, low-density updates, and anchor-centric saliency) to the observed generation-stability differences in this ablation; they do not establish an end-to-end causal guarantee.

B.10 Pass@n performance across model scales

Figures 14–16 report the original pass@n comparisons. These sample-based scores are separate from the primary pass@1 results and do not quantify uncertainty.

Table 22: Generation dysregulation rates for ablation experiments (Mistral-7B, GSM8K).

Configuration	Dysregulation Rate (%)
Ablation 1: Selection Criterion (fixed $\alpha = 1.5$)	
RKL (SCF-RKL)	0.5
Magnitude	1.0
Sign Disagreement	4.7
Random	72.3
Ablation 2: Sparsity Level (fixed RKL criterion)	
$\alpha = 0$	4.4
$\alpha = 1.0$	0.8
$\alpha = 1.5$ (default)	0.5
$\alpha = 2.0$	4.7
$\alpha = 2.5$	34.9
$\alpha = 3.0$	53.6
$\alpha = 3.5$	69.4
$\alpha = 4.0$	69.5
Source Models (reference)	
Anchor (MathCoder2)	73.4
Complement (Mistral-Ins)	0.0

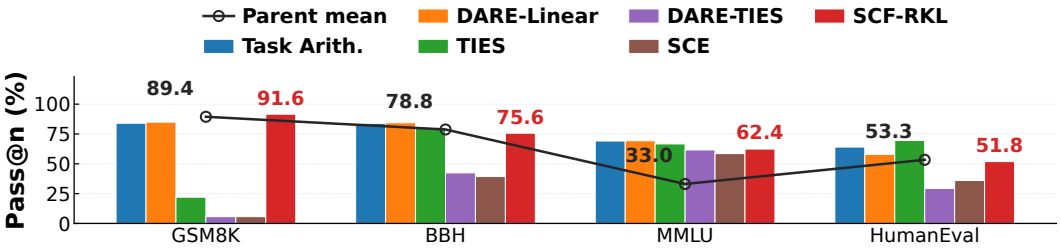


Figure 14: Pass@n scores for fused models (bars), with the mean of the two source models shown as a line on LLaMA-3-8B-Instruct–Derived Models.

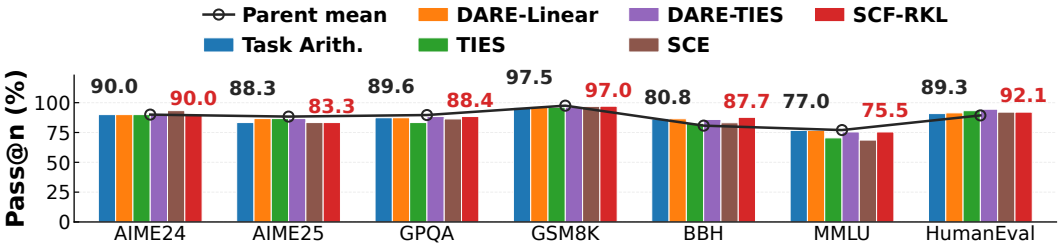


Figure 15: Pass@n scores for fused models (bars), with the mean of the two source models shown as a line on DeepSeek-R1-Distill-Qwen2.5-14B–Derived Models.

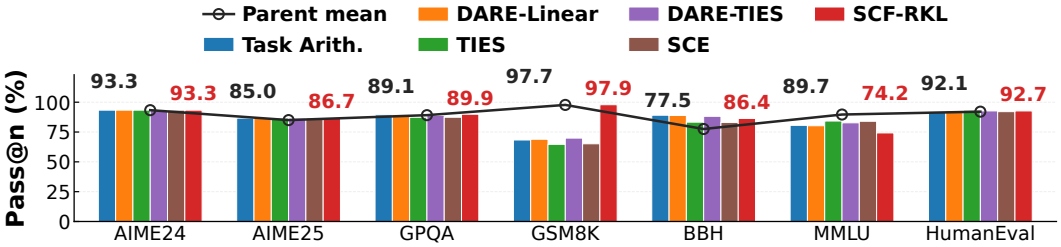


Figure 16: Pass@n scores for fused models (bars), with the mean of the two source models shown as a line on DeepSeek-R1-Distill-Qwen2.5-32B–Derived Models.

B.11 32B and 14B fusion: per-benchmark accuracy

32B scale. Table 23 shows complementary strengths: the Math parent leads on AIME25, LiveCodeBench, and MMLU, while the Science parent leads on AIME24, GPQA, and BBH. SCF-RKL reaches 62.50 on AIME25, 69.19 on GPQA, and 62.46 on LiveCodeBench, exceeding the better parent by 0.42, 0.41, and 0.81 points, respectively. Its GSM8K score is 95.20, compared with 24.39–27.39 for the evaluated merging baselines. It has the highest fused IFEval and IFBench scores, although the Math parent remains stronger on IFEval. These are task-level observations, not evidence identifying particular generative-control circuits.

Table 23: Performance of merging TinyR1-Preview-Math-32B (Math), and TinyR1-Preview-Science-32B (Science) on all datasets. Task scores follow the original evaluation protocol; Avg. is the equal mean of the displayed task scores. The best overall result (including base models) is underlined; the best among fused models is **bolded**.

Method	Advanced Reasoning				Knowledge & Reasoning			Code Gen	Instruction		Avg
	AIME24 ↑	AIME25 ↑	GPQA ↑	LCB ↑	GSM8K ↑	BBH ↑	MMLU ↑	Human Eval↑	IFEval Strict↑	IFBench Strict↑	
Math	74.58	62.08	67.99	61.65	95.42	48.18	78.10	85.52	63.42	7.76	64.47
Science	78.02	61.88	68.78	61.47	96.02	61.11	51.65	85.52	60.79	10.15	63.54
Task Arith.	**77.50**	61.25	68.18	57.80	26.57	75.26	68.56	85.98	51.62	6.27	57.90
DARE-Linear	**77.50**	61.25	68.18	55.91	26.19	**75.29**	68.84	**86.59**	50.23	8.06	57.80
TIES	74.38	61.04	67.08	50.18	24.39	57.43	73.70	85.52	52.52	9.25	55.55
DARE-TIES	76.15	60.62	67.08	56.90	27.39	73.33	71.27	85.06	55.45	8.96	58.22
SCE	74.38	61.04	67.08	49.91	24.68	57.49	**74.74**	84.45	53.51	7.46	55.47
SCF-RKL	76.98	**62.50**	**69.19**	**62.46**	**95.20**	70.31	61.17	86.28	**57.90**	**11.34**	**65.33**

14B scale. TinyR1-Preview-Math-14B and TinyR1-Preview-Code-14B encode reasoning via structurally different strategies (symbolic CoT vs. code-block inference), making this a harder fusion scenario than 32B. The reported baselines degrade AIME24 by 8–12 points and incur >99% dysregulation on GSM8K (93→71). SCF-RKL has the highest fused score on 7 of 10 benchmarks, including a LiveCodeBench gain of $+6.55$ over the strongest parent (62.46 vs. 55.91); this supports complementary transfer in the evaluated pair rather than a general emergence claim.

Table 24: Performance of merging TinyR1-Preview-Math-14B (Math), and TinyR1-Preview-Code-14B (Code) on all datasets. Task scores follow the original evaluation protocol; Avg. is the equal mean of the displayed task scores. The best overall result (including base models) is underlined; the best among fused models is **bolded**.

Method	Advanced Reasoning				Knowledge & Reasoning			Code Gen	Instruction		Avg
	AIME24 ↑	AIME25 ↑	GPQA ↑	LCB ↑	GSM8K ↑	BBH ↑	MMLU ↑	Human Eval↑	IFEval Strict↑	IFBench Strict↑	
Math	74.27	61.25	61.65	55.91	93.73	48.15	61.83	70.27	64.93	8.36	60.04
Code	72.50	58.44	60.01	55.20	93.46	69.29	68.39	84.15	43.05	5.07	60.96
Task Arith.	61.98	54.17	51.39	38.53	71.49	66.15	**65.17**	84.15	38.08	10.48	54.16
DARE-Linear	62.92	54.06	51.39	38.53	71.44	66.17	64.85	83.08	37.50	11.34	54.13
TIES	63.23	52.81	52.08	44.18	70.74	60.70	58.94	84.91	34.70	9.55	53.18
DARE-TIES	61.88	49.27	51.07	40.05	71.74	64.75	63.76	**85.67**	37.25	10.48	53.59
SCE	65.62	56.25	51.61	44.35	72.65	60.53	57.12	**85.67**	34.47	**12.24**	54.05
SCF-RKL	**73.96**	**58.02**	**56.36**	**62.46**	**93.42**	**69.92**	63.47	83.23	**42.12**	7.46	**61.04**

Notably, as shown in Figure 17, the SCF-RKL fused model surpasses both base models on all seven safety benchmarks, demonstrating effective integration of safety-critical parameters while preserving or even improving reasoning and general capabilities. These results suggest that SCF-RKL mitigates the commonly observed safety–capability trade-off and offers a principled approach to building models that are both capable and robust.

Table 25: Performance of merging Mistral-7B-Instruct-v0.2 (Base 0), MathCoder2-Mistral-7B (Base 1) on all datasets. *: The merged model fails to provide structured response. All metrics are reported as `pass@n` scores unless otherwise specified.

Merging Methods	Models	General Knowledge & Reasoning			Code Generation
		GSM8K↑	BBH↑	MMLU↑	HumanEval↑
w/o Merging	Base 0	68.23	66.36	51.99	16.46
	Base 1	45.19	73.03	33.53	0.61
Task Arithmetic	Fuse	78.70	71.61	44.14	39.02
DARE-Linear	Fuse	79.76	71.38	43.03	39.63
TIES	Fuse	44.35	72.85	33.85	0.00
DARE-TIES	Fuse	81.80	75.39	56.09	36.59
SCE	Fuse	43.67	72.41	33.76	0.00
SCF-RKL (Ours)	Fuse	82.79	74.58	56.96	41.46

Table 26: Performance of merging TinyR1-Preview-Math-32B (Math), and TinyR1-Preview-Science-32B (Sci) on all datasets. Task scores follow the original evaluation protocol; Avg. is the equal mean of the displayed task scores.

Methods	Advanced Reasoning				Gen. Know. & Reasoning			Code Gen	Inst. Following		Avg
	AIME24 ↑	AIME25 ↑	GPQA ↑	LCB ↑	GSM8K ↑	BBH ↑	MMLU ↑	Human Eval↑	IFEval Strict↑	IFBench Strict↑	
w/o Merge	74.6	62.1	68.0	61.7	95.4	48.2	78.1	85.5	63.4	7.8	64.5
	78.0	61.9	68.8	61.5	96.0	61.1	51.7	85.5	60.8	10.2	63.5
SCF-KL	77.6	63.3	68.3	59.1	88.6	61.0	73.8	86.7	58.3	8.7	64.5
SCF-RKL	77.0	62.5	69.2	62.5	95.2	70.3	61.2	86.3	57.9	11.3	65.3

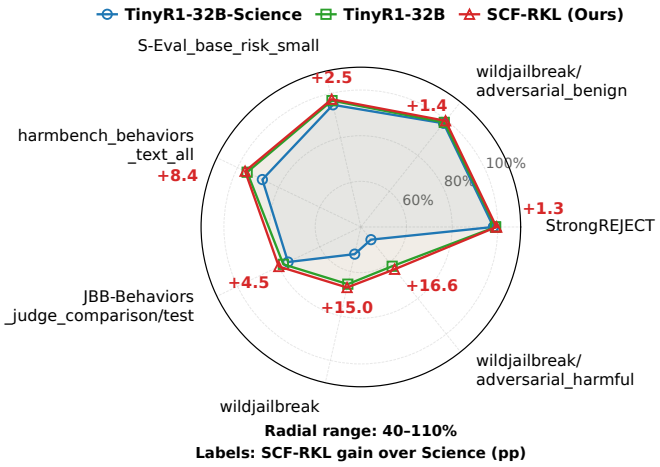


Figure 17: Safety profiles of source and fused models derived from DeepSeek-R1-Distill-Qwen2.5-32B. Numbers denote gains over TinyR1-32B-Science in percentage points.

Table 27: Computational cost of Task Arithmetic (TA) vs. SCF-RKL in GPU mode. Time is wall-clock seconds; memory is peak GPU allocation.

Scale	GPU	Time (s)		GPU Mem (GB)	
		TA	SCF-RKL	TA	SCF-RKL
7B	H800	38	59	1.46	4.15
7B	A100	30	62	1.46	4.15
14B	H800	152	331	8.71	24.7
14B	A100	215	1005	8.71	24.7
32B	H800	220	352	8.71	24.7
32B	A100	422	2145	8.71	24.7

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